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# Palaeomagnetic study of Palaeoproterozoic granitoids from the Voronezh Massif, Russia

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## SUMMARY

A palaeomagnetic investigation has been carried out of rocks from the eastern part of the Voronezh Massif, which constitutes, together with the Ukrainian Shield, the Sarmatian segment in the southern part of the East European Craton. The samples were collected in a quarry close to the town of Pavlovsk (50.4°N, 40.1°E), where a syenitic-granitic body intrudes Archaean units. U–Pb (zircon) dating has yielded an age of 2080 Ma for the intrusion.

Two characteristic magnetic components, A and B, were isolated by thermal and alternating-field demagnetization. Component A was obtained from granites and quartz syenites (11 samples) and has a mean direction of $D = 229°$, $I = 28°$, and a pole position at 12°N, 172°E. This pole is close to a contemporary mean pole (9°N, 187°E) for the Ukrainian Shield, which implies that the Voronezh Massif and the Shield constituted a single entity at 2.06 Ga. These poles differ from contemporaneous poles of the Fennoscandian Shield, indicating that the relative positions of the two shields were different from their present configuration about 2100 Myr ago.

A component B, isolated only in quartz monzonites (five samples), has a mean direction $D = 144°$, $I = 49°$, and a pole position at 4°N, 251°E, which is close to late Sveconorwegian (approximately 900 Ma) poles for Baltica. This suggests that the East European Craton was consolidated some time between 2080 and 900 Ma. Comparison with other palaeomagnetic data permit us to narrow this time span to 1770–1340 Ma.

**Key words:** East European Craton, granitoids, palaeomagnetism, Proterozoic.

## INTRODUCTION

This study is part of a larger project which aims to elucidate the ancient configuration of the Precambrian plates that form the East European Craton (EEC) using palaeomagnetic methods on well-dated rock units. Palaeomagnetic data are important in geodynamic investigations for reconstructing ancient movements and configurations of continental blocks. Apparent polar wander paths (APWPs) are time sequences of palaeomagnetic poles and define the kinematics of crustal blocks.

The 1960–70s were dominated by the concept that the EEC with its Archaean core was uniformly built up and acted as a single entity during the Proterozoic. Recently, however, that concept has been questioned, and it has been suggested that the EEC is made up of three major crustal segments: Fennoscandia (including the Baltic Shield), Volgo–Uralia and Sarmatia (see, for example, Gorbatschev & Bogdanova 1993). Sarmatia, which occupies most of the southern half of the Craton, comprises the Ukrainian Shield and the Voronezh Massif. They are separated from each other by the Palaeozoic Dniepr–Donets Aulacogen (DDA) with a deeply subsided Precambrian basement (Fig. 1).

Geological and geophysical investigations to test this three-segment model for the EEC are in progress and are being handled by various international programmes (for example EUROPROBE of the ESF/ILP, IGCP-371 Structure and Correlation of the Baltic Precambrian in NE Europe and North Atlantic Realm, COPENA).

Palaeomagnetic studies have been carried out in the Baltic Shield since the late 1950s, and an APWP for the Shield based on palaeopoles from well-dated rock units has been constructed (Elming *et al.* 1993). There are fewer data available from the rest of the EEC. However, the palaeomagnetic data from the Baltic and the Ukrainian shields pose questions concerning their relative positions during Palaeoproterozoic times. Our studies have therefore aimed to (1) establish the Palaeoproterozoic position of the Voronezh Massif relative to the Ukrainian Shield in order to test correlations across the DDA, and (2) obtain additional evidence for the Palaeoproterozoic positions of Fennoscandia and Sarmatia.

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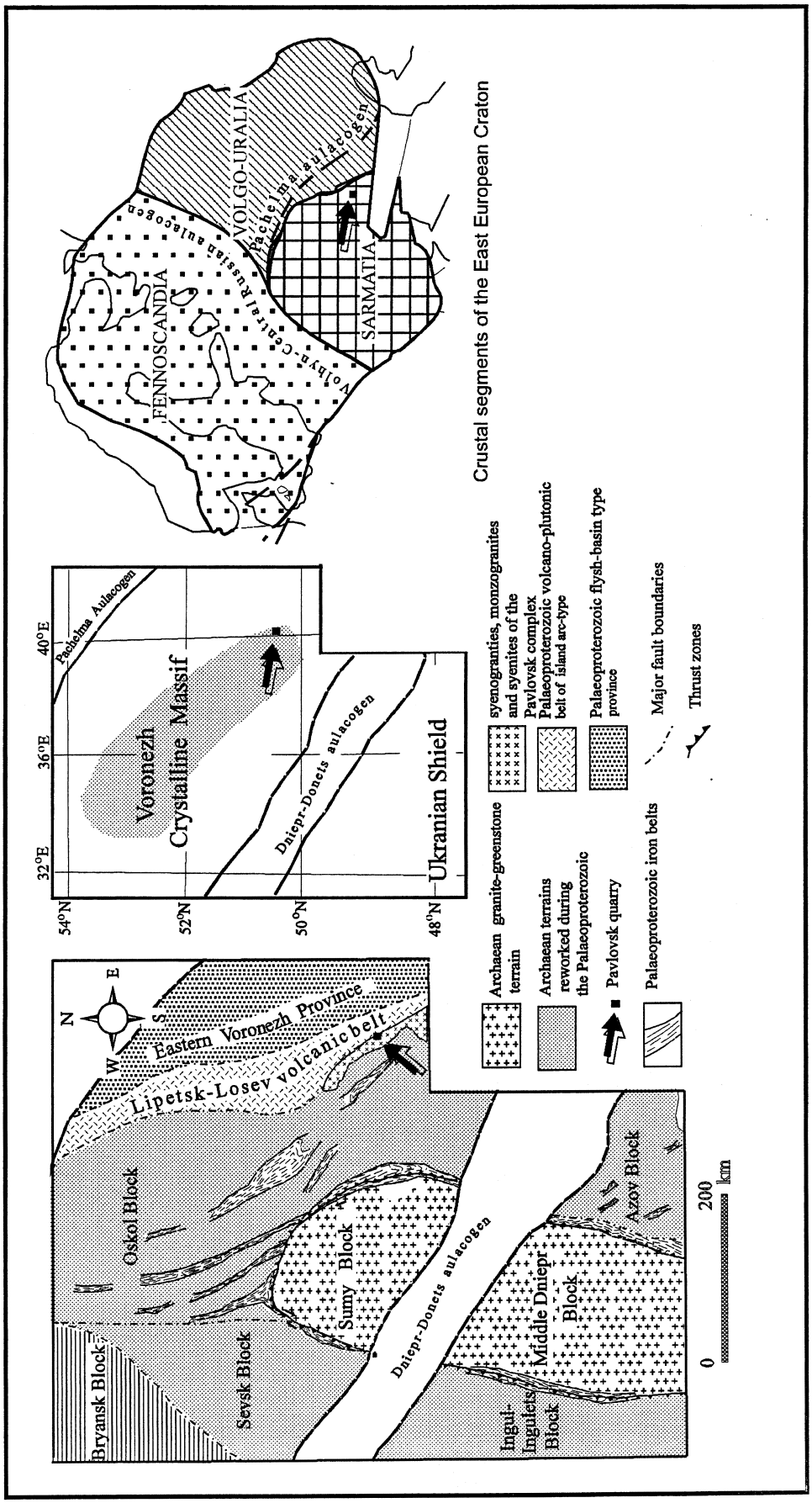



**Figure 1.** The geological position of the Pavlovsk granitoid massif in Sarmatia (after Shchipansky & Bogdanova 1996).

## GEOLOGICAL SETTING

The basement of the Voronezh Massif is buried beneath a Phanerozoic platform cover ranging from tens to thousands of metres in thickness. Thus the principal sources of information on the Precambrian geology of the Voronezh Massif are subsurface samples obtained mostly from drillings and several industrial iron-ore pits where structures can be mapped and compared with geophysical data.

In the eastern part of the Massif, natural basement outcrops of the crystalline basement occur on the shore of the Don river, near the town of Pavlovsk. These outcrops and nearby quarries expose a granitoid intrusion of the so-called Pavlovsk complex which belongs to a subalkaline igneous series. This intrusion is situated at the eastern margin of the Oskol block, a complex Archaean terrain formed from 3.3 to 2.6 Ga and reworked in the Palaeoproterozoic (Schipansky & Bogdanova 1997). The Pavlovsk intrusion is 30–50 km wide and extends for more than 200 km along the major NNW-trending fault zone (Fig. 1) that separates the Archaean part of Sarmatia from the East Voronezh accretional belt of the Palaeoproterozoic (2.1 Ga) juvenile crust. A syenite from the Pavlovsk quarry has yielded a U–Pb zircon age of $2078 \pm 4$ Ma (E. V. Bibikova, personal communication 1994). Together with other 2.1–2.0 Ga subalkaline and even alkaline intrusions, which are widespread in this part of the Voronezh Massif, the Pavlovsk intrusion marks a magmatic belt formed by convergent tectonics between the margin of the Archaean craton of Sarmatia and Palaeoproterozoic oceanic crust sited to the east in present-day coordinates.

The intrusion is composed of a variety of coarse-grained subalkaline granitoids comprising granites, adamellites, quartz syenites, quartz monzonites and monzodiorites. Numerous pegmatitic and fine-grained aplitic dykes cut the main body. Zones of deformation and faulting subdivide this intrusion into separate blocks. The granitoids in such zones are mylonitized and represented by fine-augen gneisses. Samples were collected both from undeformed granites and from fault zones.

## PALAEOMAGNETIC TECHNIQUES

17 orientated hand-samples were collected from three sites in a large quarry near the city of Pavlovsk (50.4°N, 40.1°E). Four cubes 2 cm in side and two cylinders 2.5 cm in diameter and 2.3 cm high were obtained from each hand-sample. Measurements were carried out at the palaeomagnetic laboratory of VNIGRI, St Petersburg, Russia and at the Department of Geology, Lund University, Sweden.

The natural remanent magnetization (NRM) was measured on JR-4 (St Petersburg) and JR-5 (Lund) spinner magnetometers. A CS-2 heating device connected with a KLY-2 Kappabridge (Lund) was used for measuring the temperature dependence of the magnetic susceptibility. For thermal stepwise demagnetization to 600–650 °C, a laboratory-built oven with residual fields near 10 nT was used in St Petersburg, while a Schoenstedt TSD-1 furnace was used in Lund. KLY-2 Kappabridge susceptibility meters were used at both laboratories to check for possible changes in magnetic mineralogy due to heating. Stepwise alternating-field (AF) demagnetization (Lund) was applied up to peak fields of 80 mT.

Orthogonal plots (Zijderveld 1967), stereographic diagrams and principal component analysis (Kirschvink 1980) were used to analyse and separate the magnetization vectors. Fisher (1953) statistics were used for calculating the mean directions.

## PALAEOMAGNETIC RESULTS AND ANALYSIS

NRM intensities before demagnetization range from 1.5 to 2000 mA $m^{-1}$, and magnetic susceptibility from 0.1 to $55 \times 10^{-3}$ SI. The susceptibility versus temperature measurements are presented in Fig. 2. These curves make it possible to determine the Curie temperature ($T_C$) of the magnetic minerals in the rocks.

In Fig. 2, samples P2, P9 and P10 indicate the presence of magnetite, together with a mineral with a $T_C$ of around 400 °C, while sample P16 indicates the presence of magnetite and

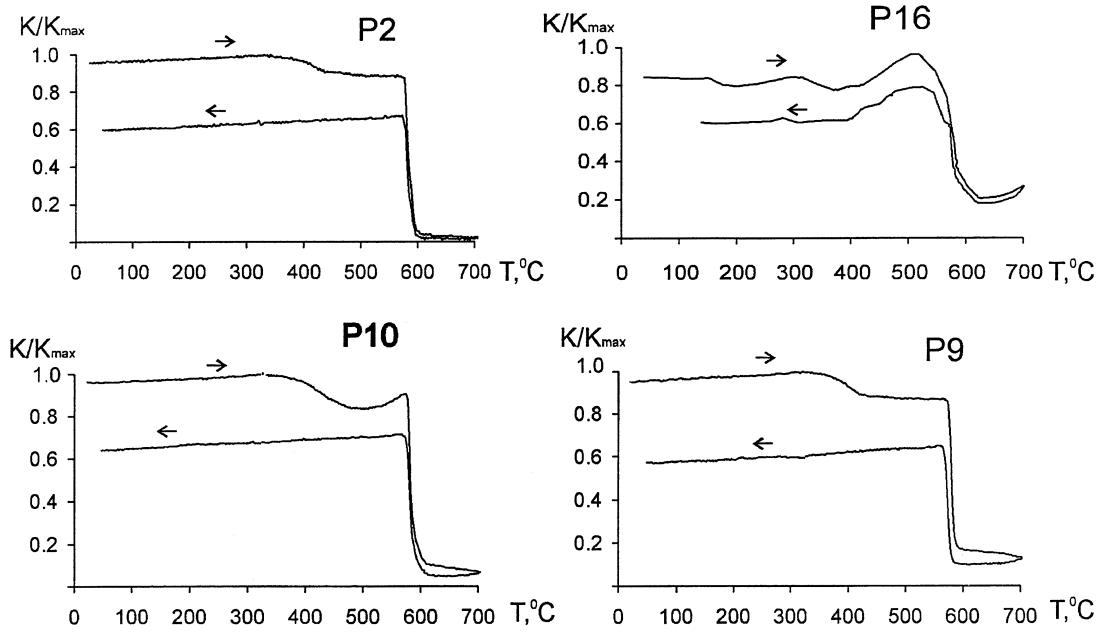



**Figure 2.** Typical curves of the temperature dependence of magnetic susceptibility.

haematite only as carriers of the magnetization. Susceptibility versus temperature results, measured after each heating–cooling step, are presented in Figs 3 and 4. Two types of curves were obtained: specimens P2–1 L, P4–2 L, P9S1 and P10S1 all show a decrease in $K$ above 400 °C of 25–40 per cent, while no decrease was observed in specimen P16S2. This is a further indication that haematite was formed during heating. In sample P16 the changes in NRM became irregular

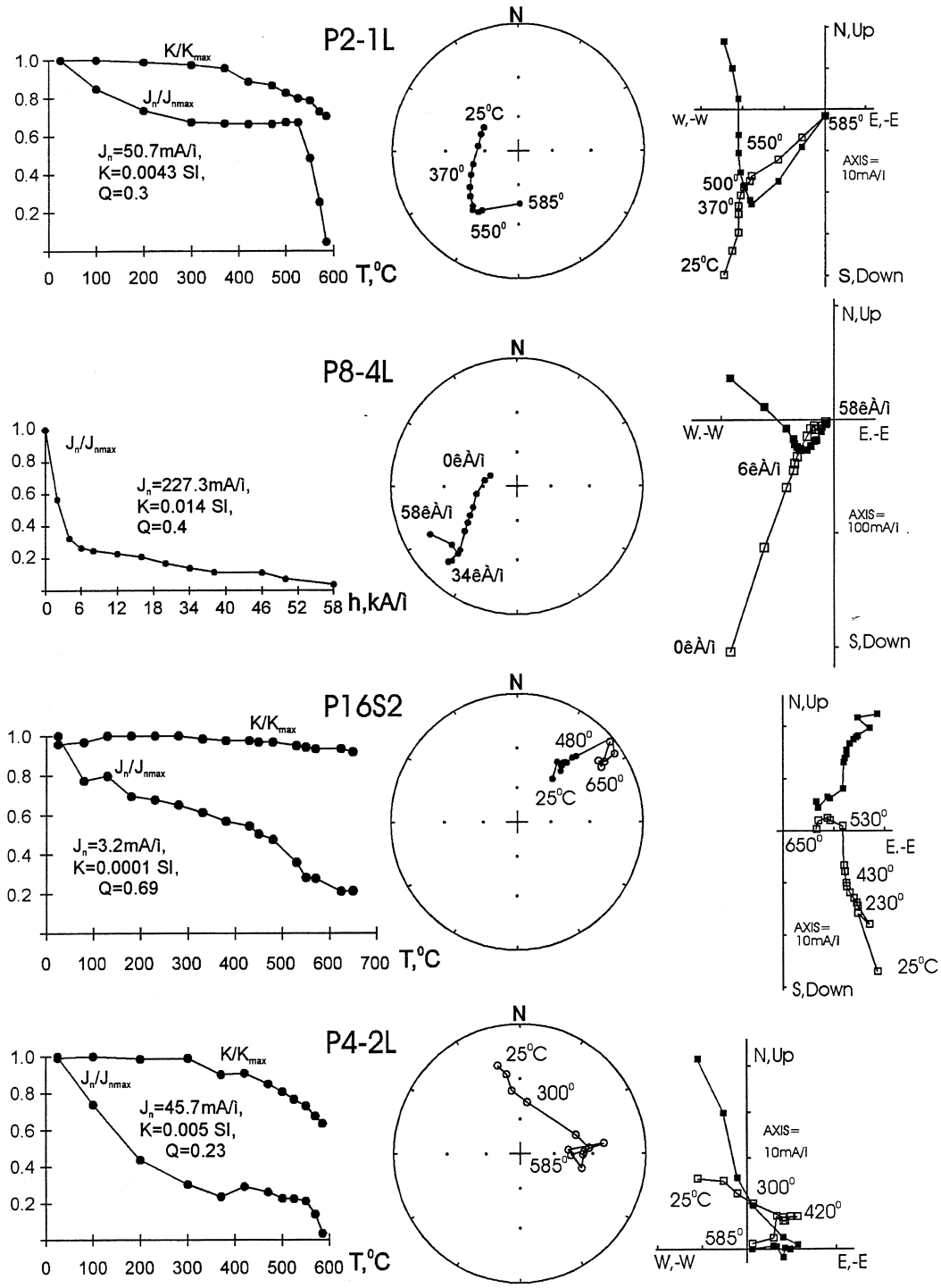



**Figure 3.** Typical demagnetization results for granite (samples P2–1I, P4–2 L, P8–4 L) and quartz syenite (sample P16S2) from the Pavlovsk quarry; component A. Left: NRM-decay curves and susceptibility measurements after each heating–cooling cycle. Values for NRM-decay curves and susceptibility measurements after heating–cooling cycles are given in normalized units. Centre: equal-area stereographic projections. Closed (open) symbols on stereographic projections represent projections on the lower (upper) hemisphere. Right: orthogonal projections (Zijderveld diagrams) of the NRM vector end-points for specimens of rocks during thermal and AF demagnetization treatments. For all orthogonal projections, closed (open) symbols represent projection onto the horizontal (vertical) plane, where N is North, E is East, and Up is vertical up. On the Zijderveld diagrams, axis scales (mA $m^{-1}$) are shown. Values of the temperatures (°C) and alternating field (kA $M^{-1}$) are shown.

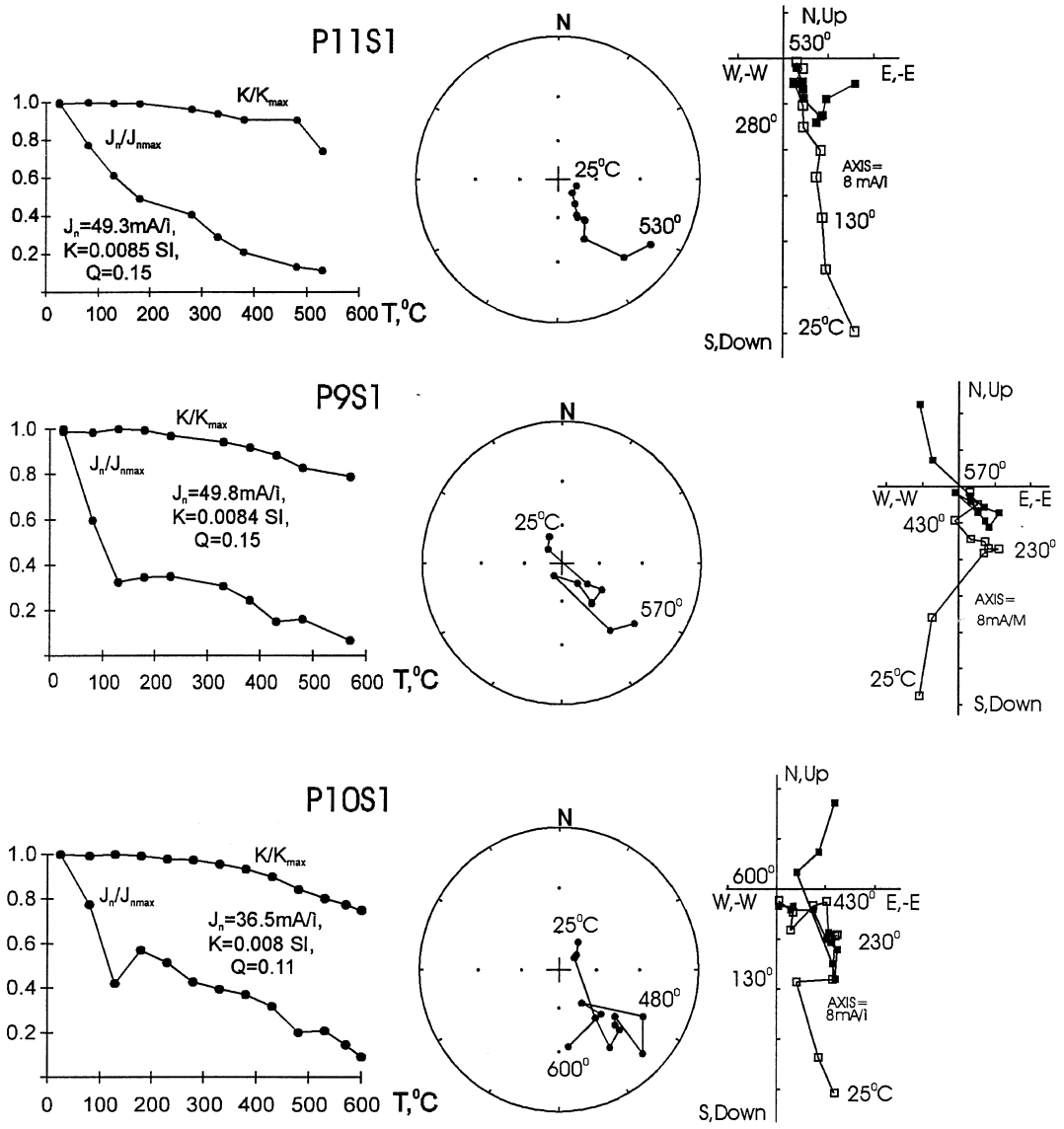



**Figure 4.** Typical demagnetization results of granite and dolerite from the Pavlovsk quarry; component B. Conventions as in Fig. 3.

above 650 °C and it was not possible to define haematite in this sample. The blocking temperature spectra correspond to magnetite in most samples, and only in sample P16 do we have a mixture of magnetite and haematite.

Pure magmatic magnetite and secondary haematite have been observed in thin sections. Microprobe analyses did not show any sign of Ti in the samples studied. The results obtained are presented in Table 1 and Figs 3, 4 and 5.

Two characteristic components, A and B, were isolated during the AF and thermal demagnetization tests. In addition, a component which is probably of recent origin and acquired in the Earth's present field was removed at temperatures below 200–400 °C and peak AF fields below 12 kA m$^{-1}$. Component A, $D = 229°$, $I = 28°$ (Fig. 3, P2–1 L, P8–4 L), is carried by magnetite and removed by 600 °C and by 50–80 kA M$^{-1}$.

Some samples gave an anti-parallel negative direction (Fig. 3, P4–21, P16S2) carried by magnetite and probably haematite and removed at 600–650 °C. In some cases, the magnetization became erratic after heating to 550 °C or AF demagnetization to 40 kA m$^{-1}$, although about 10 per cent of the magnetization remained in the specimens. Above 550 °C, the susceptibility of some samples started to increase slowly, probably due to the destruction of thermally unstable minerals and the formation of magnetite during the heating–cooling process.

The directions after demagnetization are presented in Fig. 5. In Fig. 5(a) all directions are given, and in Fig. 5(b) the directions with negative inclination are reversed. The negative and positive group mean directions of the populations shown in Fig. 5(b) are anti-parallel and pass a reversal test, according to class C of McFadden & McElhinny (1990).

The magnetization of samples with negative inclination is carried by magnetite (Fig. 3, specimen P4–2 L) and probably haematite (Fig. 3, specimen P16S2), indicating that these minerals became magnetized at roughly the same time. Mineralogical analysis suggests that all samples with component A contain primary magmatic magnetite, but the Koenigsberger ratios for

**Table 1.** Palaeomagnetic mean directions and pole positions for the Voronezh Massif.

| | *n* | Decl. (°) | Incl. (°) | $\alpha_{95}$ (°) | k | Palaeolat. (°) | Plat. (°N) | Plong. (°E) | $d_p$ (°) | $d_m$ (°) |
|---|---|---|---|---|---|---|---|---|---|---|
| A. Granite, syenite | 30 | 229 | 28 | 6.2 | 15 | 15 | 12 | 172 | 3.7 | 6.8 |
| B. Quartzmonzonite, deformed | 13 | 144 | 49 | 8.2 | 23 | 30 | 4 | 251 | 7.1 | 10.8 |

Palaeopoles used for comparison, from Elming *et al.* (1993)

| | N | $A_{95}$ (°) | Plat. (°N) | Plong. (°E) |
|---|---|---|---|---|
| Baltic Shield *ca.* 2120 Ma | 2 | – | 37 | 302 |
| " " *ca* 1880 Ma | 15 | 3.5 | 43 | 237 |
| " " *ca.* 845 Ma | 3 | 7.9 | 8 | 250 |
| Ukrainian Shield *ca.* 2060 Ma | 3 | 22 | 9 | 187 |

*n*: number of specimens used for calculations; Decl. and Incl.: mean declination and inclination after demagnetization; $\alpha_{95}$: semi-angle of the 95 per cent cone of confidence. k: best estimate of Fisher's (1953) precision parameter; Plat.: pole latitude; Plong.: Pole longitude; $d_p$ and $d_m$: half-axes in the 95 per cent ovals of confidence; N: number of poles used; $A_{95}$: half-angle of the 95 per cent circle of confidence of the mean pole.

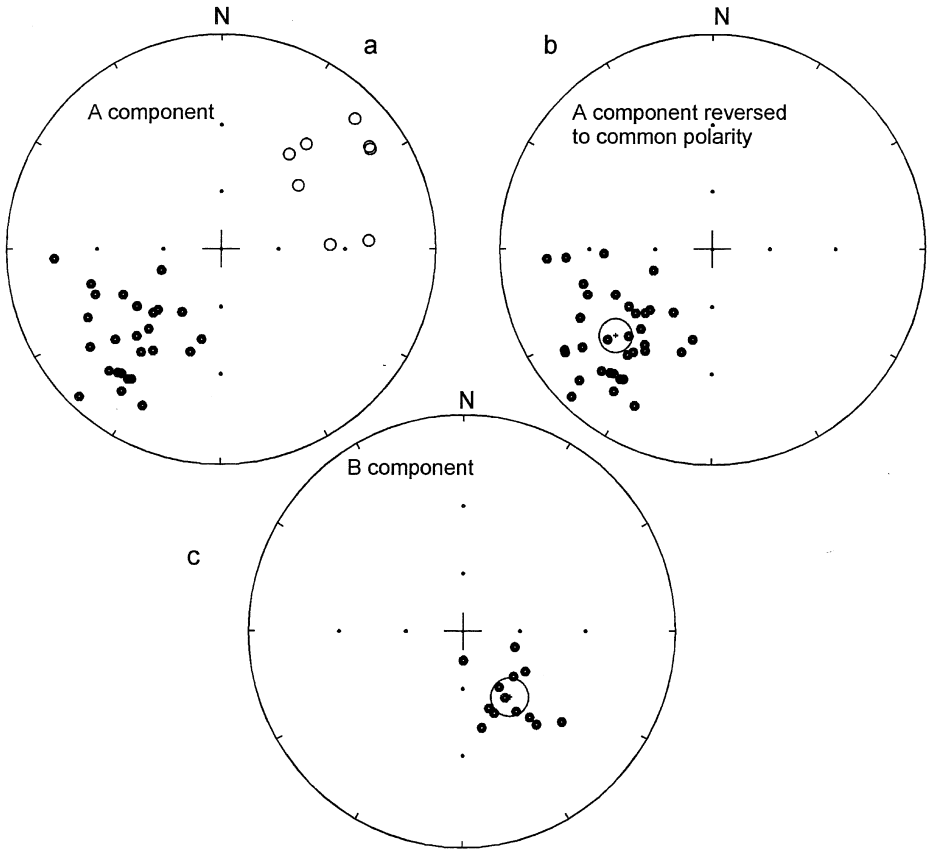



**Figure 5.** Stereograms with directions obtained after demagnetization: (a) characteristic components, group A; (b) same with negative components reversed; (c) mean directions, component B. Filled (open) circles denote the lower (upper) hemisphere. Small crosses and circles denote mean directions and $\alpha_{95}$ confidence circles.

all samples are less than 1.0 suggesting that the NRM was imparted by autometamorphic transformations during rock formation. Thus component A probably has an age of around 2078 Ma.

Component B was obtained in samples from an aplitic dyke approximately 300 m from the site where the single A-component rocks were collected. The rocks in the dyke have been deformed, fractured and affected by hydrothermal reworking. In these samples the susceptibility decreased steadily during treatment. This may be due to the destruction of primary magmatic magnetite and the formation of secondary haematite during the heating and oxidation above 400 °C (Fig. 2), and may be the cause of the poor stability of the remanence.

The B component has mean declination = 144°, inclination = 49° (Fig. 4c). Its blocking temperature is close to 600 °C. The age of this component may be expected to be younger than the age of component A due to the reworking of the magmatic granite in a fault zone.

## DISCUSSION

The data from the Voronezh Massif can be used for correlation with corresponding data from the Ukrainian and Baltic shields (Table 1). The 2078 ± 4 Ma A-pole is situated close to a 2060 Ma pole for the Ukrainian Shield presented by Elming *et al.* (1993), which indicates that the Voronezh Massif and the Ukrainian Shield at that time constituted a single crustal entity. By comparison with available contemporary data from the Baltic Shield, it is clear that the relative orientation of the two shields was different from their present orientation. The palaeolatitude for the Voronezh Massif was 14° at 2080 Ma. Present data from the Baltic Shield yield palaeolatitudes between 32° and 22° during the period 2120–1880 Ma (Elming *et al.* 1993).

The B component yields a pole at 4°N, 251°E. This is in the vicinity of some Sveconorwegian poles for the Baltic Shield situated at 8°N, 250°E, and indicates that these shields were in approximately the same relative position during the period 1.25–0.8 Ga. In most parts of the East European Craton, this period (Riphean) was a time of very intense rifting of the consolidated crust, mainly along the Palaeoproterozoic boundaries between Fennoscandia, Sarmatia and Volgo–Uralia (Bogdanova *et al.* 1996). The Pavlovsk rocks that yielded pole B are strongly cataclased and occur along zones of faulting, possibly connected with rifting processes in the Riphean.

In Fig. 6, the relative positions of Sarmatia (the Ukrainian Shield and the Voronezh Massif) and the Baltic Shield are given based on new and previously published palaeomagnetic data.

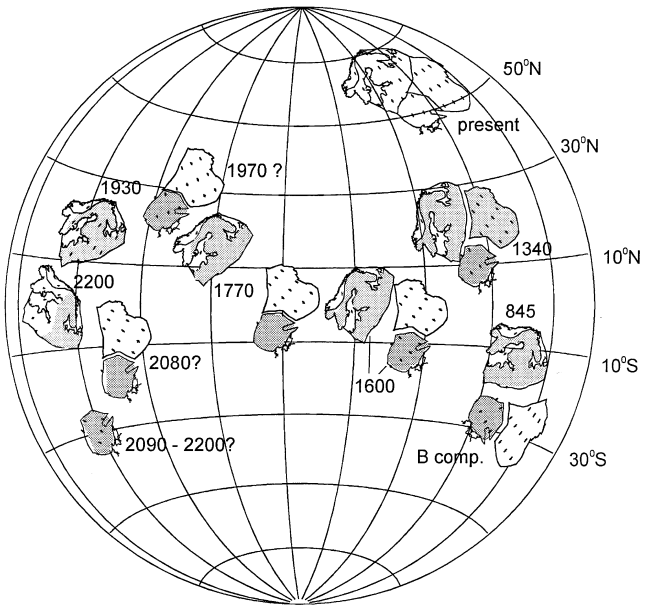



**Figure 6.** The latitudinal positions of the Fennoscandian, Volgo–Uralian and Sarmatian segments between 2200 and 845 Ma according to Table 2. Consolidated crust of segments is shown by shaded areas.

Owing to the lack of reliable data for the period 2100–1400 Ma, the Volgo–Uralian segment is not included. As mentioned above, however, it appears to have taken the same position as Sarmatia since 2.0 Ga. The Fennoscandian segment was situated between 40° and 10° in latitude and rotated anticlockwise by about 90° between 2100 and 1345 Ma (Elming *et al.* 1993). The latitudinal displacement was about 10°. The Sarmatian segment had a latitudinal displacement of approximately 25° but no significant rotation occurred during the above period.

The Voronezh Massif is in the Pavlovsk quarry area covered by flat-lying (1°–3° of tilt) Devonian sedimentary rocks, and the Pavlovsk intrusion belongs to a post-orogenic event (Shchipansky & Bogdanova 1996). There is therefore no reason to make tectonic adjustments to the palaeomagnetic directions.

The interpretation of Precambrian palaeomagnetic data is problematic. According to geological data, no significant displacements have occurred along the Archaean–Palaeoproterozoic boundaries after the Proterozoic (Shchipansky & Bogdanova 1996), making it possible to construct an APWP unique to Sarmatia since these times. The preservation of a 2080 Ma palaeomagnetic remanence may have occurred because Volgo–Uralia and Sarmatia had already collided at approximately 2.0 Ga (Bogdanova *et al.* 1996), whilst the Pavlovsk Massif, being an internal part of the Volgo-Uralia-Sarmatia continent, did not experience tectonic and compositional reworking until the Riphean rifting took place.

Regarding polarity, we use the Sarmatian and Fennoscandian palaeopoles situated in the recent western hemisphere as north poles and the corresponding NRM inclination as normal. This convention has been used in Table 2. However, the Vendian–Permian APWP for the East European Craton (Torsvik & Meert 1995) implies that the Vendian pole situated in Siberia is a south pole. If this is true and if the Neoproterozoic track of the APWP (Elming *et al.* 1993) is correct, the poles that form the Early Proterozoic part of the APWP are also south poles. To settle the polarity problem it is necessary to obtain well-defined palaeomagnetic and radiometric data for each tectonic unit of the craton.

## CONCLUSIONS

The palaeomagnetic study of the Pavlovsk intrusion yielded a reliable palaeomagnetic pole for *ca.* 2080 Ma at 12°N, 172°E, which is close to the 2060 Ma mean pole for the Ukrainian Shield located at 9°S, 187°E (Elming *et al.* 1993). This implies that the Voronezh Massif was in the same position relative to the Ukrainian Shield as at present. However, comparison with results from the Baltic Shield from rocks of about the same age suggests that these two shields were then in positions different from their present configuration.

The palaeolatitude for the Ukrainian Shield and the Voronezh Massif was 14°S at 2080 Ma, while the centre of the Baltic Shield changed in palaeolatitude from 32°N to 22°N during the period 2120 Ma to 1880 Ma (Elming *et al.* 1993).

The B-component pole, 4°N, 251°E, is situated close to poles from a group of late Sveconorwegian dolerites, the Blekinge–Dalarna dolerites, with an age of approximately 900 Ma (Pisarevsky & Bylund 1998), and poles with an age of 845 Ma (Elming *et al.* 1993). This indicates that these shields were then in approximately the same relative positions as at present.

**Table 2.** Some palaeomagnetic poles for crustal segments of the East Europen craton.

| N | Segment | Age Ma | Method | Plat. (°N) | Plong. (°E) | $d_p$ (°) | $d_m$ (°) | $A_{95}$ (°) | Reference |
|---|---|---|---|---|---|---|---|---|---|
| | Sarmatia | | | | | | | | |
| 1** | | 800–900? | B comp. | 4 | 251 | 7 | 10 | | This study |
| 2* | | 1340 | K-Ar | 41.6 | 170.5 | 5.5 | 10.6 | | Mikhailova *et al.* (1983, 1989) |
| 3* | | 1600 | K-Ar | 19.0 | 172.0 | 1.7 | 3.2 | | Mikhailova (1982) |
| 4* | | 1770 | U-Pb | 26.0 | 173.7 | 2.1 | 4.1 | | Mikhailova *et al.* (1994) |
| 5* | | 1970? | U-Pb | 57.4 | 168.6 | 3.8 | 6.3 | | Iosifidi *et al.* (1996) |
| 6 | | 2060 | K-Ar | 9.0 | 187.0 | | | 22 | Elming *et al.* (1993) |
| 7* | | 2078 | U-Pb | 12.0 | 172.0 | 3.8 | 6.3 | | This study |
| 8* | | 2090–2200 | K-Ar | −2.0 | 172.0 | | | 8.0 | Mikhailova & Kravchenko (1987) |
| | Fennoscandia | | | | | | | | |
| 9** | | 845 | | 8.0 | 250.0 | | | 7.9 | Elming *et al.* (1993) |
| 10** | | 1340 | | 45.0 | 169.3 | | | | Elming *et al.* (1993) |
| 11* | | 1600 | | 28 | 192 | | | 7.6 | Elming *et al* (1993) |
| 12 | | 1770 | U-Pb | 37.8 | 210.6 | 7.7 | 4.0 | 5.5 | Damm *et al.* (1997) |
| 13 | | 1770 | | 45.1 | 206.4 | 6.5 | 3.6 | 4.8 | Damm *et al.* (1997) |
| 14 | | 1770 | U-Pb | 40.5 | 229.1 | | | 8.1 | Fedotova *et al.* (1999) |
| 15* | | 1770 | mean | 42 | 215 | | | 14 | |
| 16* | | 1930 | U-Pb | 40 | 247 | 5 | 8 | | Mertanen & Pesonen (1994) |
| 17* | | 2120 | | 37 | 302 | | | | Elming *et al.* (1993) |
| 18 | | 2200 | U-Pb | −0.8 | 245.4 | 11.6 | 7.2 | 9.1 | Damm *et al.* (1997) |
| 19 | | 2200 | | 12.9 | 314.1 | | | 8.6 | Fedotova *et al.* (1999) |
| 20* | | 2200 | mean | 7 | 280 | | | | |
| | Volgo-Uralia | | | | | | | | |
| 21* | | 1340 | K-Ar | 44.0 | 171.0 | 4.0 | 7.0 | | Danukalov (1979) |

*Poles used for reconstruction, Fig. 6. **Poles for Sarmatia and Fennoscandia taken from the Fennoscandian APWP (Elming *et al.* 1993). Other conventions as in Table 1.

Geological data (Bogdanova 1996) indicate that the East European Craton was consolidated by 1850–1700 Ma. Palaeomagnetic data imply that this consolidation occurred after 1770 but not later than 1340 Ma. The lack of palaeomagnetic and isotopic age data for the period between 1800 and 1340 Ma makes it difficult to determine when they were assembled, but our data do not contradict the geological evidence.

## ACKNOWLEDGMENTS

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